

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and operating large-scale distributed systems. Currently builds platform infrastructure at YouTube supporting billions of daily requests. Previously led reliability engineering at Whatnot serving 5M+ daily active users. Built developer platforms at Google spanning 9 years across Ads and Cloud organizations. Expert in **Go**, **Python**, **C++**, and **Kubernetes**. Reduced live playback incidents by 80% at YouTube through edge caching redesign. Managed on-call for systems serving 50K+ concurrent viewers. Seeking senior infrastructure or platform engineering roles requiring end-to-end ownership and operational depth.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, Python

Platform Infrastructure: Kubernetes, Terraform, Docker, CI/CD, Service Mesh, Linux

Distributed Systems: Kafka, gRPC, Redis, Thrift, Load Balancing, Caching

Cloud Platforms: GCP, AWS, Azure Storage, Spanner, Bigtable

Observability: Prometheus, Grafana, Distributed Tracing, On-Call, SRE

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Led platform infrastructure team responsible for live streaming playback serving billions of daily requests. Redesigned edge caching architecture reducing latency by 60% using **Kubernetes** failover and **Redis** hot path optimization.
- Built real-time analytics pipeline processing 10M+ events per second with **Kafka** and **Flume**. Reduced processing latency from 5 minutes to 2 minutes for Live dashboard queries.
- Improved Live playback reliability from 95% to 99.9% uptime through multi-region failover redesign. Implemented automated rollback reducing incident blast radius by 80%.
- Designed and deployed **gRPC** API gateway handling 100K+ requests per second for creator analytics. Reduced p99 latency from 500ms to 100ms through connection pooling.
- Established on-call rotation covering 3 services with **Prometheus** alerting and **Grafana** dashboards. Decreased mean time to resolution from 45 minutes to 12 minutes.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Designed real-time bidding system for live auctions using **Go** and **Redis**. Reduced bid submission latency from 100ms to 10ms p99 through optimized serialization.
- Built **Kubernetes**-based auto-scaling system handling traffic spikes of 10x baseline during peak sales events. Reduced infrastructure costs by 35% through dynamic resource allocation.
- Led reliability engineering for live commerce platform serving 5M+ daily active users. Achieved 99.99% uptime for auction processing pipeline handling \$50M+ monthly GMV.
- Managed on-call rotation for platform services with **PagerDuty** integration. Reduced alert fatigue by 60% through intelligent grouping and runbook automation.
- Implemented distributed tracing with **OpenTelemetry** across 30+ microservices. Identified 15 performance bottlenecks reducing overall latency by 40%.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Built internal developer platform serving 50K+ Google engineers using **Kubernetes** and **Terraform**. Reduced deployment pipeline time by 75% and cut infrastructure provisioning from days to minutes.
- Created distributed tracing system analyzing 10TB+ of **Spanner** and **Bigtable** data daily. Enabled engineers to debug latency issues across 200+ services without code changes.
- Designed data pipeline processing 1PB+ daily for cross-product analytics using **C++** and **Bigtable**. Reduced query latency from 10 seconds to 200 milliseconds for critical dashboards.

- Established SRE practices for 50+ internal platform services. Implemented error budgets and reliability SLOs reducing P0 incidents by 70% across the organization.

Software Engineer | Google | 02/2011 - 02/2015

- Developed core infrastructure services for Google Ads serving 1M+ queries per second. Optimized **C++** hot paths improving throughput by 25% and reducing CPU usage significantly.
- Participated in on-call rotation for critical Ads infrastructure with **Prometheus** alerting. Resolved 20+ severity-1 incidents with documented root causes and preventive measures.
- Built configuration management system managing 1M+ service configurations across 10 data centers. Implemented **MySQL** sharding strategy reducing lookup latency by 50%.
- Designed monitoring aggregation service collecting 1B+ metrics daily. Migrated from monolithic architecture to **Python** microservices reducing deployment coupling.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Contributed to Azure Storage distributed storage system serving 10K+ internal users with **C++** and **Windows Azure** stack. Implemented consistency checking algorithms reducing data corruption incidents by 30%.
- Built automated testing framework increasing code coverage from 60% to 90%. Reduced manual QA cycles from 2 weeks to 3 days through continuous integration pipelines.
- Optimized build verification system processing 5K+ builds daily. Decreased average build time by 40% through parallel execution and incremental compilation strategies.
- Participated in design reviews for storage quorum protocols. Authored technical documentation for 3 core system components used by 50+ engineers.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed internal tooling for test case management using **C#** and **SQL Server**. Automated regression testing reducing manual effort by 25% for the Visual Studio team.
- Created monitoring dashboards for build health tracking. Presented project outcomes to 20+ engineers demonstrating 15% improvement in build failure detection time.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007